

Lecture 11 Review Handout

Correlation and Repeated Measures | Statistics 367-1-4361

Lecture Overview

Lecture 11 moves from regression models, where one variable is treated as explanatory, to settings where variables or repeated measurements are jointly structured. The first half develops correlation as a population parameter inside a bivariate normal likelihood and compares correlations across two groups. The second half uses repeated wine ratings to separate latent wine quality from judge-specific generosity and discrimination. The lecture then translates the same modeling ideas into fixed/random effects, frequentist F-tests, and LOO-based model comparison.

11.1 — The Correlation Coefficient

Correlation is introduced as a symmetric, non-causal relationship between two noisy measurements. Unlike regression, neither variable has to be treated as the controlled or explanatory variable. The goal is to move from the sample correlation coefficient to a population correlation parameter that can appear inside a likelihood.

New Definitions and Terms

- **Correlation** — a symmetric association between two variables, without assigning one as the cause or predictor of the other.
- **Population correlation coefficient** — the parameter ρ describing the direction and strength of linear association in the population.
- **Sample correlation coefficient** — the estimate r , also called Pearson's r or the product-moment correlation.
- **Covariance matrix** — a matrix with variances on the diagonal and covariances on the off-diagonal.

- **Multivariate normal likelihood** — a likelihood for vector-valued observations, with means, variances, and covariances modeled jointly.

Formulas

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

The population correlation is normalized covariance. Its sign gives direction and its magnitude gives strength.

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

The sample correlation r estimates ρ . In the normal model it is the maximum-likelihood estimate.

$$= \begin{pmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix}$$

The covariance matrix can be built from the two standard deviations and the correlation.

Figure

Correlation is symmetric

- Auditory and visual reaction times
 - Axes can be swapped
 - Both variables are measures with noise

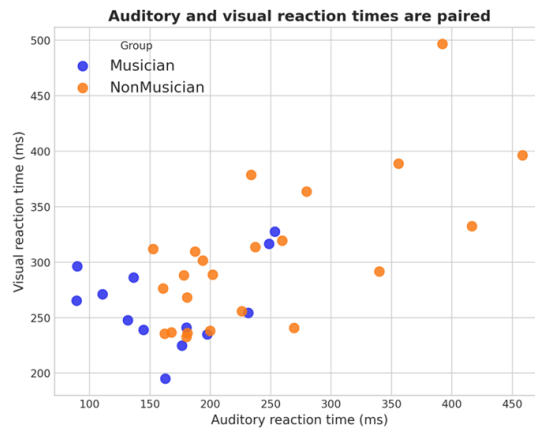


Figure 1: Slide 5 — Correlation is symmetric

How do we model correlations?

- $$\begin{pmatrix} x \\ y \end{pmatrix} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$
- $$\boldsymbol{\mu} = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix} \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{pmatrix} \quad \sigma_{xy} = \sigma_x \sigma_y \rho$$
- We want to estimate ρ .
 - For Bayesian inference, ρ must appear inside a likelihood.
 - We need a likelihood that describes a pair of outcomes jointly
 - Each participant contributes a vector observation
 - (auditory RT, visual RT)
 - The bivariate normal distribution gives us this
 - Our first multivariate likelihood

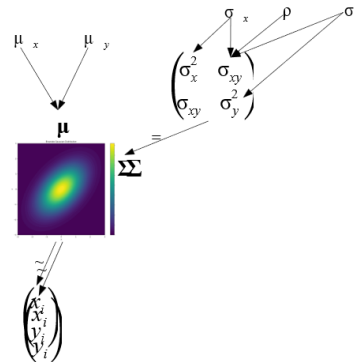


Figure 2: Slide 12 — How do we model correlations?

11.2 — Two-group Bivariate Normal Model

The audiovisual reaction-time data are modeled as two-dimensional observations: auditory and visual reaction time for each subject. The model estimates separate means, standard deviations, and correlations for musicians and non-musicians. The group difference in correlation is calculated inside the model as a deterministic parameter.

New Definitions and Terms

- **Vector observation** — one data point containing multiple measured outcomes, here (auditory RT, visual RT).
- **Group-specific correlation** — a separate ρ_g parameter for each group.
- **Scaled Beta prior for correlation** — a Beta prior on $\theta_\rho \in [0, 1]$, transformed to $\rho \in [-1, 1]$.
- **Deterministic contrast** — a parameter calculated from other model parameters, such as $\Delta\rho$.

Formulas

$$\theta_{\rho,g} \sim \text{Beta}(\alpha, \beta), \quad \rho_g = 2\theta_{\rho,g} - 1$$

The Beta prior is convenient because it is bounded; the transformation maps it to the valid correlation range.

$$\mathbf{y}_i \sim \text{MvNormal}(\boldsymbol{\mu}_{g_i}, g_i)$$

Each subject contributes a two-variable response vector, and the group index selects that subject's mean vector and covariance matrix.

$$\Delta\rho = \rho_{\text{Musician}} - \rho_{\text{NonMusician}}$$

The group contrast is part of the posterior, not a separate post-hoc calculation.

Figure

The graphical version of the model is more compact

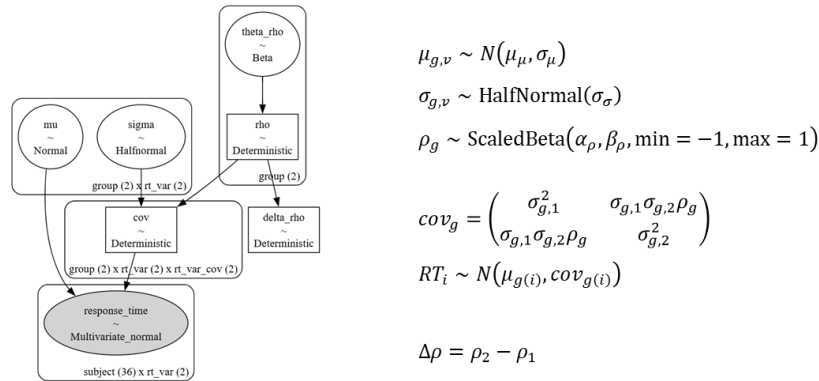


Figure 3: Slide 22 — The graphical version of the model is more compact

$\Delta\rho$: the posterior for the group difference

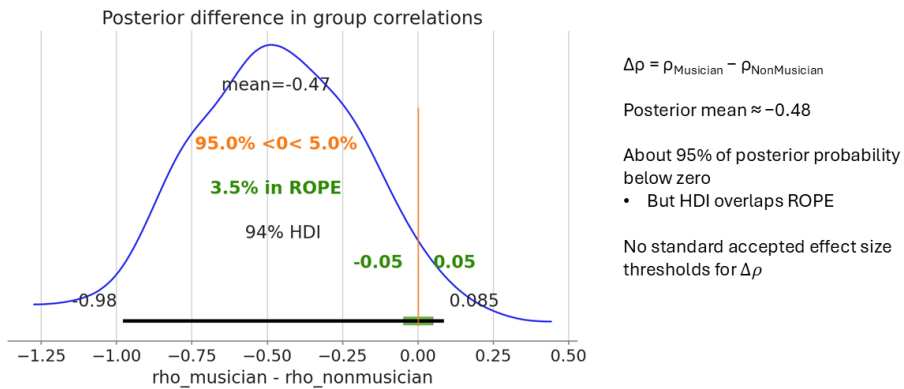


Figure 4: Slide 24 - posterior for the group difference in correlation

11.3 — Frequentist Translation

The frequentist translation uses Pearson's r as the point estimate and Fisher's z transform to construct confidence intervals and compare independent correlations. The Bayesian model estimates the same kind of population quantity, but represents uncertainty with posterior distributions rather than sampling intervals and null tests.

New Definitions and Terms

- **Fisher-z transform** — the transformation $\text{arctanh}(r)$, used because transformed correlations are approximately normal.
- **Fisher-z confidence interval** — a confidence interval computed on the Fisher-z scale and then transformed back to the correlation scale.
- **Fisher-z difference test** — a frequentist test for whether two independent population correlations differ.

Formulas

$$z = \text{arctanh}(r) = \frac{1}{2} \log \left(\frac{1+r}{1-r} \right)$$

Fisher's transform maps correlations from $[-1, 1]$ to the real line.

$$z \pm 1.96 \sqrt{\frac{1}{n-3}}$$

This gives an approximate 95% confidence interval on the Fisher-z scale; applying $\tanh(\cdot)$ returns the interval to the correlation scale.

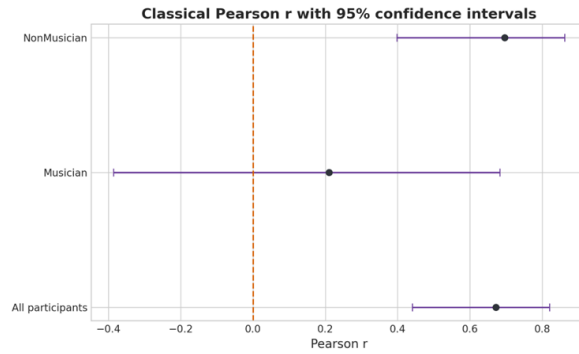
$$z_{\Delta} = \frac{z_1 - z_2}{\sqrt{\frac{1}{n_1-3} + \frac{1}{n_2-3}}}$$

This is the standard test statistic for comparing two independent correlations.

Figure

Pearson r and Fisher-z confidence intervals

$$r = \hat{\rho}_{MLE} \quad CI: \tanh\left(\operatorname{arctanh}(r) \pm \frac{1.96}{\sqrt{n-3}}\right)$$



Pearson r:

- The standard frequentist estimate of ρ

Fisher-z:

- transform to $\operatorname{arctanh}(r)$,
- Compute Normal CI
- Back-transform

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Figure 5: Slide 26 — Pearson r and Fisher-z confidence intervals

Comparing two independent correlations

$$Z = \frac{\operatorname{arctanh}(r_1) - \operatorname{arctanh}(r_2)}{\sqrt{\frac{1}{n_1 - 3} + \frac{1}{n_2 - 3}}}$$

- Frequentist question
 - Are ρ_1 and ρ_2 significantly different?
- Standard test
 - Fisher-z difference test

	sample	n	pearson_r	ci 2.5%	ci 97.5%	p_value_H0_rho_0
0	All participants	36	0.672	0.441	0.819	0.00
1	Musician	13	0.211	-0.385	0.682	0.49
2	NonMusician	23	0.696	0.398	0.861	0.00

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Figure 6: Slide 27 — Comparing two independent correlations

11.4 — Repeated Measurements: Wine Ratings

The wine ratings data introduce repeated measurements: every judge rates every wine. A first model ignores judge identity and lets scores vary only by wine. This creates a simple latent wine-quality model, with wine effects represented as deviations from the grand mean.

New Definitions and Terms

- **Repeated measurements** — multiple observations sharing a structured source, here many ratings from the same judges and many ratings of the same wines.
- **Zero-sum constraint** — a constraint that effects sum to zero, so each level is interpreted as a deviation from the grand mean.
- **ZeroSumNormal** — a PyMC distribution for normally distributed effects constrained to sum to zero.

Formulas

$$\mu_{ij} = \alpha + Q_i, \quad \sum_i Q_i = 0$$

In the first wine model, the expected score depends only on wine identity. The wine-quality effects Q_i are deviations from the grand mean.

Figure

9 judges, 20 wines, 180 ratings: every judge scores every wine

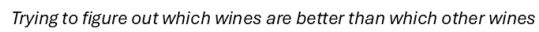


Figure 7: Slide 30 — The Wines2012 data: wines and judges

Pooled wine quality (ignoring judges)

Wine	Mean Quality (approx.)	Quality Range (approx.)
B2	0.55	0.00 to 1.00
J2	0.30	0.00 to 0.90
D2	0.25	0.00 to 0.90
D1	0.20	0.00 to 0.90
B1	0.25	0.00 to 0.90
A1	0.15	0.00 to 0.70
E2	0.15	0.00 to 0.70
A2	0.10	0.00 to 0.60
E1	0.10	0.00 to 0.60
G2	0.05	0.00 to 0.50
F1	0.05	0.00 to 0.50
F2	0.05	0.00 to 0.50
C1	-0.10	-0.20 to 0.40
G1	-0.10	-0.20 to 0.40
I1	-0.15	-0.25 to 0.35
J1	-0.20	-0.30 to 0.30
H2	-0.20	-0.30 to 0.30
H1	-0.25	-0.35 to 0.25
C2	-0.40	-0.50 to 0.10
I2	-0.80	-1.00 to -0.60

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11.5 — Wine Quality and Judge Generosity

The heatmap shows systematic row differences: some judges score higher or lower overall even when the wine ordering is fixed. The new model adds judge generosity as a hierarchical source of variation. Wine effects remain non-hierarchical because the goal is to estimate the specific wines, while judge effects share information through a population-level standard deviation.

New Definitions and Terms

- **Judge generosity** — a judge-specific level shift; generous judges give higher scores on average and harsh judges give lower scores on average.
- **Hierarchical judge effect** — judge effects are estimated with a shared variance parameter, allowing information sharing across judges.
- σ_{judge} — the population standard deviation of judge generosity.

Formulas

$$\mu_{ij} = \alpha + Q_i + G_j$$

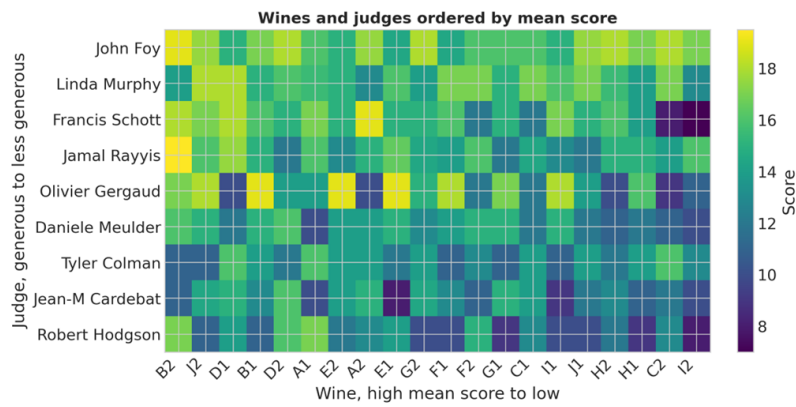
Wine score is modeled as the sum of a grand mean, a wine-quality effect, and a judge-generosity effect.

$$G_j \sim \text{ZeroSumNormal}(\sigma_{\text{judge}})$$

The spread of judge generosity is learned from the data. If the posterior for σ_{judge} is away from zero, judges differ meaningfully in their average scoring levels.

Figure

The heatmap reveals judge generosity

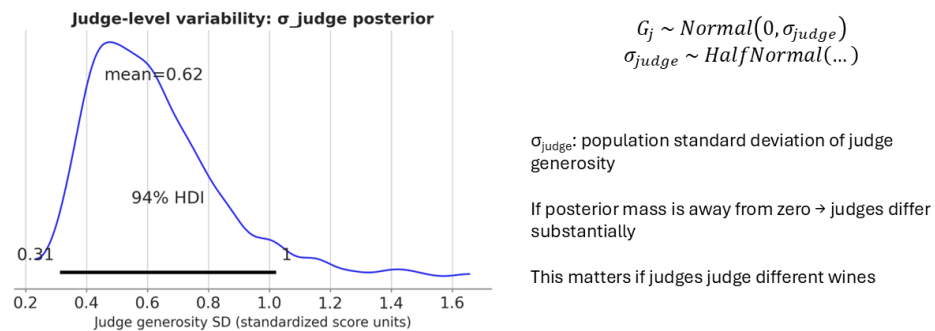


Rows differ systematically even after wines are ordered.

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Figure 9: Slide 35 — The heatmap reveals judge generosity

Differences between judges could matter



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Figure 10: Slide 39 — Differences between judges could matter

11.6 — Fixed and Random Effects

Fixed and random effects are introduced as frequentist terminology that can be translated into Bayesian model structure. In the Bayesian translation, fixed effects are level-specific parameters with fixed priors, while random effects are level-specific parameters governed by a hierarchical variance. The wine model treats wines as fixed-effect-like because the levels matter, and judges as random-effect-like because the spread of judge effects is the main target.

New Definitions and Terms

- **Fixed effect** — an effect where the levels themselves matter and are estimated directly.
- **Random effect** — an effect where the levels are treated as replaceable, and the population-level spread is estimated.
- **Varying intercept** — a group-specific intercept, written in formula notation as $(1 \mid \text{group})$.
- **Wilkinson notation** — the formula grammar used by `Bambi`, `lme4`, `nlme`, and related modeling tools.

Formulas

```
score_z ~ 0 + wine + (1 | judge)
```

This formula means: one wine-specific term for each wine identity, plus a varying intercept for judge.

Wilkinson notation for random effects

```
score_z ~ 0 + wine + (1 | judge)
```

- $0 + \text{wine}$: one wine-specific term per wine identity (fixed-effect-like)
- $(1 | \text{judge})$: varying intercept per judge; Bambi estimates judge-level spread from data
- Bambi recovers the same wine quality estimates as the explicit PyMC model
- Same notation used in frequentist lme4 / nlme: shared grammar across frameworks

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Figure 11: Slide 42 — Wilkinson notation for random effects

11.7 — Frequentist Multi-factor Models

The lecture then translates the wine structure into ordinary least squares and ANOVA language. A model with wine and judge factors is fit with OLS, and the ANOVA table partitions the fit into terms. The full-model F-test asks whether the model as a whole explains variation, while a partial F-test asks whether adding one factor improves fit relative to a reduced model.

New Definitions and Terms

- **Multi-factor model** — a regression model with more than one categorical or continuous term.
- **Dummy variable** — a coded regression variable representing one level or contrast of a categorical factor.
- **Full-model F-test** — a test of whether the full regression model improves fit relative to an intercept-only model.
- **Partial F-test** — a test of whether a factor improves fit relative to a reduced model without that factor.

Formulas

Let SS_{full} be the residual sum of squares for the full model and p the number of fitted regression coefficients.

$$F_{\text{full}} = \frac{(SS_{\text{null}} - SS_{\text{full}})/(p - 1)}{SS_{\text{full}}/(n - p)}$$

The numerator measures improvement over the null model per model degree of freedom. The denominator measures residual variance per residual degree of freedom.

$$F_{\text{factor}} = \frac{(SS_{\text{reduced}} - SS_{\text{full}})/(p_{\text{full}} - p_{\text{reduced}})}{SS_{\text{full}}/(n - p_{\text{full}})}$$

The partial F-test asks whether the additional factor reduces residual sum of squares enough to justify its degrees of freedom. In the wine example, 20 wines correspond to 19 dummy variables and 9 judges correspond to 8 dummy variables.

Figure

Frequentist multi-factor models

- We've seen models like these before
- Fit with “ordinary least squares”
- Produce an “ANOVA table”

```
wine_ols_formula = "score_z ~ wine + judge"
wine_ols_fit = smf.ols(wine_ols_formula, data=wines).fit()
```

	model	F	df_model	df_resid	p_value	R2	adj_R2
0	score_z ~ wine + judge	3.522	27.0	152.0	0.0	0.3849	0.2756

	SS	df	MS	F	p_value
wine	19.5585	19.0	1.0294	1.4210	0.1245
judge	49.3296	8.0	6.1662	8.5119	0.0000
Residual	110.1118	152.0	0.7244	NaN	NaN

Figure 12: Slide 46 — Frequentist multi-factor models

Calculating F for a factor

- The F statistic distributes with the F distribution
 - Under the null hypothesis that the factor has no effect
- l is the number of dummy variables in the factor
- $\hat{y}_{factor}$ is the predicted value for y without the factor

$$F = \frac{MS_{factor}}{MS_{res}} = \frac{\frac{SS_{factor}}{df_{factor}}}{\frac{SS_{res}}{df_{res}}} \sim F(df_{factor}, df_{res})$$

$$SS_{factor} = \sum_{i=1}^N (\hat{y}_{factor,i} - \hat{y}_i)^2$$

$$SS_{res} = \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$df_{factor} = l$$

$$df_{res} = N - (k + 1)$$

20 wines : 19 dummy variables
9 judges: 8 dummy variables

Figure 13: Slide 48 — Calculating F for a factor

11.8 — Discrimination: Motivation and Model

Judge generosity is only a level shift. The discrimination model adds the idea that judges may differ in how strongly their scores reflect latent wine quality. A judge with higher discrimination gives scores that vary more strongly with inferred wine quality, while a low-discrimination judge compresses quality differences.

New Definitions and Terms

- **Discrimination** — judge-specific sensitivity to latent wine quality; the strength of the quality-score relationship.
- **Latent wine quality** — wine quality inferred inside the model rather than treated as an observed predictor.
- **Log-discrimination** — a log-scale parameter used so discrimination remains positive after exponentiation.
- **Population ellipse** — the fitted population distribution of judge effects, distinct from individual judge posterior uncertainty.

Formulas

$$\mu_{ij} = G_j + D_j Q_i$$

The expected score is shifted by judge generosity G_j and scaled by judge discrimination D_j .

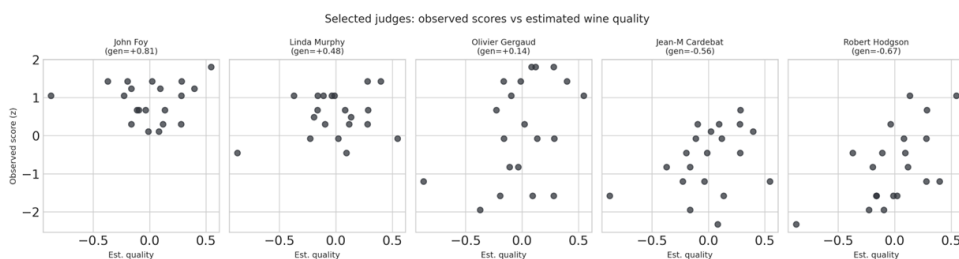
$$D_j = \exp(L_j), \quad (G_j, L_j) \sim \text{MvNormal}(\boldsymbol{\mu}_{\text{judge}}, \boldsymbol{\Sigma}_{\text{judge}})$$

Generosity and log-discrimination are modeled jointly, so their population correlation can be estimated.

Figure

Some judges separate wine quality more clearly than others

Each panel is one judge; a stronger relationship between estimated quality and score means higher discrimination.



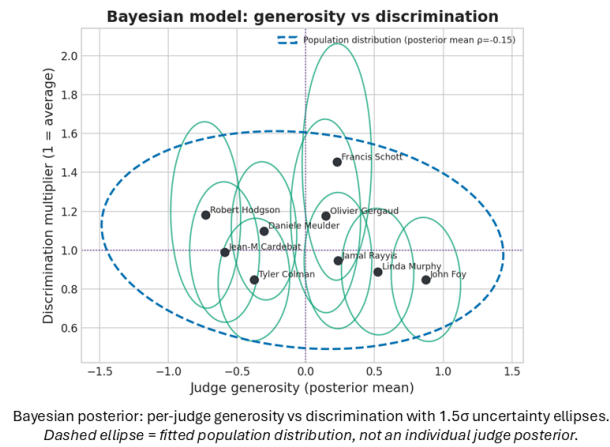
Generosity is a level shift. Discrimination is the strength of the quality-score relationship.

This motivates letting judges vary in sensitivity, not just in average score.

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Figure 14: Slide 50 — Some judges separate wine quality more clearly than others

Generosity vs discrimination: exploratory to Bayesian



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Figure 15: Slide 54 — Generosity vs discrimination: exploratory to Bayesian

11.9 — Three-model Comparison

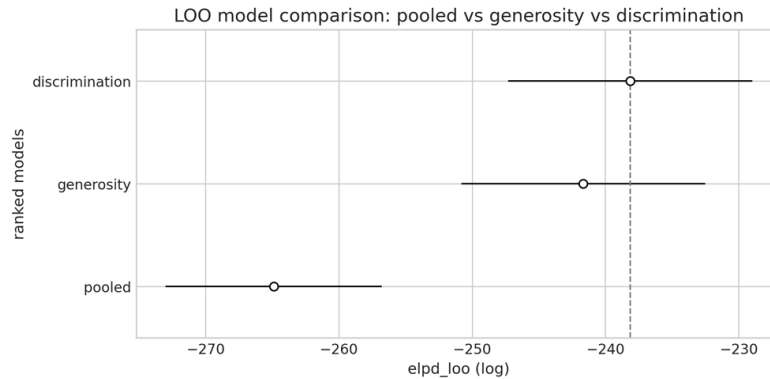
The final model comparison asks whether added complexity improves predictive performance. The pooled model allows scores to vary only by wine. The generosity model adds judge-specific intercepts, and the discrimination model adds judge-specific sensitivity to latent quality. LOO favors the generosity model over the pooled model, while the improvement from discrimination is small or uncertain.

New Definitions and Terms

- **Pooled wine model** — scores vary by wine only, with no judge structure.
- **Generosity model** — wine quality plus judge-specific average scoring level.
- **Discrimination model** — wine quality plus judge-specific average scoring level and judge-specific sensitivity to quality.

Figure

LOO results: which model predicts best?



Generosity model clearly outperforms pooled — judges differ meaningfully.
Discrimination adds little or uncertain improvement over generosity — LOO intervals overlap.

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Figure 16: Slide 57 — LOO results: which model predicts best?

11.10 — Next Time

The lecture closes by returning to the Bayesian workflow. By this point, model specification, sampling, diagnostics, posterior predictive checks, interpretation, and model comparison have become a repeated pattern. The next lecture shifts attention upstream: before modeling begins, data can already mislead through measurement problems, sampling bias, missingness, bias, and confounding.

Figure

Before the model, the data can already mislead

So far we have mostly assumed well-collected, representative data.

Target population → sample → measurements → observed data → model

What can go wrong upstream:

- Measurement: are we measuring what we think we are?
- Sampling: does the sample represent the target population?
- Missingness: are missing data informative (MNAR)?
- Bias and confounding: treatment effects versus selection effects

Next time: how analyses can be led astray before or during modeling.

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Figure 17: Slide 62 — Before the model, the data can already mislead